Drop, Reflect and Reuse a Quantum Temporary

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We characterize safe drops by contextual equivalence. Without measurement, a safe drop means the temporary ends in the same pure state at every input, so at most one reflection restores it. The criterion extends to records and loans and checks reuse in our OpenQASM 3 compiler.

1 Specifying a safe drop

When a temporary qubit is no longer needed, we want to reuse it without disturbing the rest of the computation. The standard technique, uncomputation, runs the computation that produced a temporary backwards, which can take as many gates as the computation itself [14, §1]. Whether a temporary may be safely discarded is a separate question, which languages and tools such as Silq and Unqomp settle with conservative rules for automatic reclamation that can reject safe temporaries [2, §4.4], [14, §6.2].

Consider p below, which allocates its only temporary `t`, of dimension d, in pure state ψ, runs a measurement-free computation M on `(t,v)`, and drops `t`. We compare p just before the drop with an independent witness $\mathcal{S}(P,U)$, a comparison program that uses P to prepare `t` without using the input `v`, and applies U separately to `v`:

$$p(\mathtt{v}) = \texttt{new t : [d] @ } \psi \texttt{; M; drop t; return v}$$

$$\mathcal{S}(P,U)(\mathtt{v}) = \texttt{new t : [d] @ } \psi \texttt{; P t; U v; drop t; return v}$$

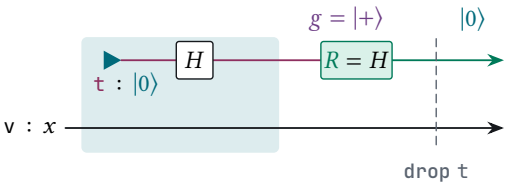
We call the drop *safe* if $p \equiv \mathcal{S}(P,U)$ for one witness on all inputs, meaning that no context can distinguish the programs stopped before their drops, with `t` and `v` returned.

We show that at a safe drop, `t` ends in one fixed pure state g at every input. From g we construct a return gate R that restores ψ on `t` alone with at most one reflection. The two examples below take a qubit `v`, with inserted return gates shown in green. To check each drop, we use the shaded *exposed prefix* before the chosen `drop` *site* (⫞). Its map K sends input x to the joint state there, with `t` first. A wire label `v : x` gives the register and its input state. Triangles (▶) mark allocation and ground symbols (⏚) end scope.

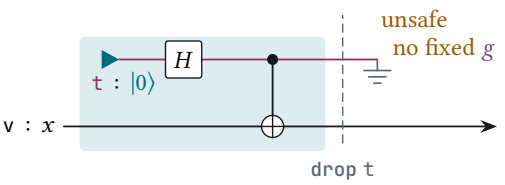
```
prog a (v : [2]) : [2] {
  new t : [2] @ |0>; H t;
  drop t; return v }
```

```
prog b (v : [2]) : [2] {
  new t : [2] @ |0>; H t; CX t v;
  drop t; return v }
```

$K^{(a)}x = |+\rangle \otimes x$



$K^{(b)}x = (|0\rangle \otimes x + |1\rangle \otimes Xx)/\sqrt{2}$



In our language, `drop` ends `t`'s scope without changing the joint state. Program (a) leaves `t` in $|+\rangle$ at every input, so H alone restores `t` to $|0\rangle$. In program (b), `CX t v` flips `v` when `t` is $|1\rangle$, giving the Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$ when `v` starts in $|0\rangle$. Both temporaries have the same lexical scope, so checking scopes would accept both drops. Silq and Unqomp's

`qfree` rules exclude H because it takes basis states to superpositions, and reject both drops. Unqomp also rejects `H t; H t`, where `t` ends in $|0\rangle$ [14, §6.2].

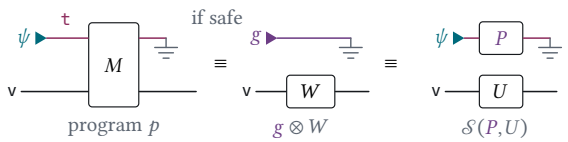
2 Checking the equivalence

$$K^{(a)} = \frac{1}{\sqrt{2}} \overbrace{\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}}^{H \otimes I} \cdot \overbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}}^{|0\rangle \otimes I} = \overbrace{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}^{K^{(a)}} \begin{matrix} K_0 \\ K_1 \end{matrix} \quad \text{slices } I/\sqrt{2},\ I/\sqrt{2}$$

$$\overbrace{\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}}^{K^{(b)} = \mathrm{CX}\,K^{(a)}} \begin{matrix} K_0 \\ K_1 \end{matrix} \quad \text{slices } I/\sqrt{2},\ X/\sqrt{2}$$

To read the joint state from the temporary's side before the drop, we group the exposed prefix's matrix rows by `t`'s basis value. For program (a), allocation $|0\rangle \otimes I$ acts before $H \otimes I$. The blue *slice* K_0 gives $|0\rangle \otimes K_0 x$ and the gold K_1 gives $|1\rangle \otimes K_1 x$. The joint state is the sum of these two terms.

A *context* is a closed program whose hole takes `v` to `(t,v)`. It supplies an input, possibly entangled with a reference, applies gates and reads all registers, recording outcome weights, as in




```
new v : [2] @ |0>; (t,v) = { □ }; CX t v; H t; return (t,v).
```

This context undoes program (b)'s gates and reads 00 with weight 1. An independent witness leaves `t` and `v` uncorrelated, so the same context reads 00 at most half the time. The witness prepares `t` in a fixed pure state g without using the input, while U computes a map W on the rest, sending input x to their joint state Wx. The witness therefore has slice $g_i W$ at basis value i, where g_i is the coefficient of $|i\rangle$ in g. For p to be equivalent to the witness, no context can distinguish their joint states before the drop, on any input. This requires the same slice form $K_i = g_i W$ for p, giving $Kx = g \otimes Wx$. In program (a), equal slices give $g = |+\rangle$ and $W = I$, while in program (b) no scalar turns I into X.

The witness prepares the dropped registers together as one *block*, without the input, and allocates matching registers for p's other temporaries so contexts can compare the full joint state, including hidden registers.

Theorem 2.1 (without measurement). *Let K be the exposed map before dropping allocated temporaries with pure allocation state ψ in a well-typed, measurement-free program p. These are equivalent.*

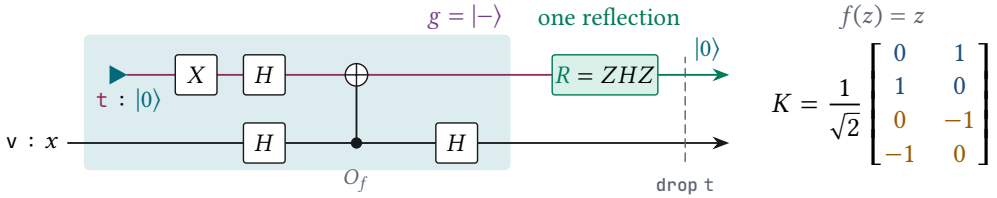
(1) $p \equiv \mathcal{S}(P,U)$ for some independent witness.

(2) $K = g \otimes W$ for a unit vector g on the block and an isometry W, a length-preserving map from the input to all other registers.

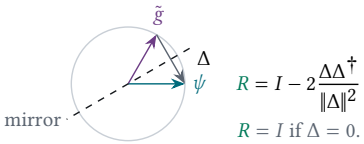
(3) One unitary on the block alone restores ψ at every input.

Deutsch–Jozsa shows why slices need only be proportional. Its oracle O_f flips `t` when $f(z) = 1$. In the minus state $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, a flip only negates `t`, so the oracle leaves

it in $|-\rangle$ and gives the data state $|z\rangle$ a sign $(-1)^{f(z)}$. These signs define $D_f|z\rangle = (-1)^{f(z)}|z\rangle$. For one data qubit, the Hadamards give $W = HD_fH$, pictured for $f(z) = z$. The slices are $W/\sqrt{2}$ and $-W/\sqrt{2}$, proportional for every f, with $g = |-\rangle$. Reversing the preparation takes H then X on `t`. The reflection $R = ZHZ$ restores `t` in one gate, and in this case both methods leave the data unchanged.



To build R, choose a phase representative $\tilde{g}$ of our *certificate* g with real, nonnegative overlap with ψ, moving the phase into W. The difference $\Delta = \psi - \tilde{g}$ points from $\tilde{g}$ to ψ. If they coincide, no gate is needed. Otherwise the Householder reflection through the mirror orthogonal to Δ sends $\tilde{g}$ to ψ [8]. The compiler aligns $\Delta/\|\Delta\|$ with the usual allocation state $|0\cdots0\rangle$, flips only that state's sign and undoes the alignment. Its QASM circuit may use several gates, but depends on the block's certificate rather than the producer's length. If $\Delta/\|\Delta\|$ is the uniform superposition of the block's basis states, then $-R$ is Grover's diffusion.



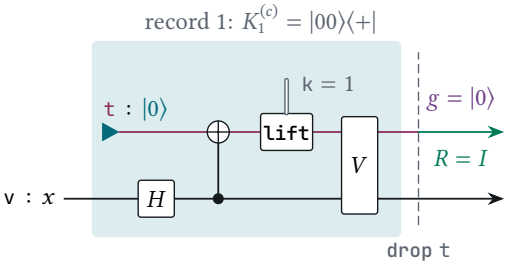
3 One measurement record at a time

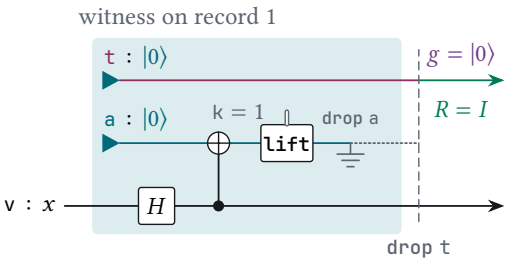
With measurement, we check each outcome sequence, or *record* o, separately. Our `lift[S] t` keeps `t` live and binds 1 for membership in a basis set S, or 0 otherwise. Only classical `if` guards read this outcome, as in dynamic lifting [6]. Thus `lift[{0}] t` returns 1 when `t` is read as 0. The exposed map K_o stays unnormalized, so $\|K_o x\|^2$ is the record's probability at a normalized input x.

Program (c) uses a two-qubit unitary V that fixes $|00\rangle$ and sends $|11\rangle$ to $(|01\rangle + |10\rangle)/\sqrt{2}$. Its independent witness measures a fresh helper qubit `a` instead of `t`:

```
prog c (v : [2]) : [2] {
  new t : [2] @ |0>; H v; CX v t;
  k = lift[{0}] t; V t v;
  drop t; return v }
```

```
prog witness (v : [2]) : [2] {
  new t : [2] @ |0>; new a : [2] @ |0>;
  H v; CX v a; k = lift[{0}] a;
  drop a; drop t; return v }
```





At input $x = \alpha|0\rangle + \beta|1\rangle$, measurement followed by V leaves these unnormalized states:

- Record 1 (safe) leaves $(\alpha + \beta)|00\rangle/\sqrt{2}$, with weight $|\alpha + \beta|^2/2$. The temporary remains in $|0\rangle$, so $R = I$.
- Record 0 (unsafe) leaves $(\alpha - \beta)(|01\rangle + |10\rangle)/2$, with weight $|\alpha - \beta|^2/2$. Its nonzero outputs are Bell pairs, so this record has no pure certificate g.

On record 1, $K_1^{(c)} = |0\rangle\otimes W_1$ with $W_1 = |0\rangle\langle+|$. The map W_1 keeps only the $|+\rangle$ component of the input, so it never lengthens a vector and is a *contraction*. For example, W_1 gives $|0\rangle$ with weight 1 at $x = |+\rangle$, but $|0\rangle/\sqrt{2}$ with weight $1/2$ at $x = |0\rangle$. Normalizing would give $|0\rangle$ in both cases, hiding the dependence of record 1's weight on the input.

If we allocate the witness's `t` and `a` in $|0\rangle$ and apply `H v; CX v a`, then these steps preserve total weight, so reproducing record 1 requires measurement. They leave `t` in $|0\rangle$, while `v` and `a` have the same value in each basis term. Measuring the independent `t` would give the same probability at every input, so the witness measures `a`. Outcome 1 leaves `t` and `a` in $|0\rangle$ and `v` in $W_1 x$, with record 1's weight. Program (c)'s unsafe record 0 still prevents reuse.

The same construction realizes any contraction W_o by giving the helper's other outcome the missing weight $\|x\|^2 - \|W_o x\|^2$. With fresh registers matching input and output spaces, the two outcomes preserve norm and extend to a unitary U [17, Thm. 2]. One `lift` then selects $W_o x$, while P prepares g from ψ on the block.

We call record o *safe* and write $p \equiv_o \mathcal{S}(P,U)$ if no context distinguishes K_o from the witness's map $g \otimes W_o$ on kept outcome 1 before the drop, omitting only its fixed $|0\rangle$ helper. Both maps stay unnormalized to retain record probabilities, whatever the other records do.

Theorem 3.1 (on a selected record). *Let K_o be the exposed map before dropping allocated temporaries on a record o of a well-typed program p. These are equivalent.*

(1) $p \equiv_o \mathcal{S}(P,U)$ for some independent witness using one `lift`.
(2) $K_o = g \otimes W_o$ for a unit vector g on the block and a contraction W_o to all other registers.
(3) One unitary on the block alone restores its pure allocation state ψ at every input on o.

If $K_o = 0$, then the record has probability zero at every input and all three conditions hold. Return gates may differ between records.

Since reuse must preserve observable behavior, we might hope to decide safety from the *summed channel*, the mixture of returned-register and reference states over records [20, §2.3.2]. With `t` prepared in $|+\rangle$, measuring `t` in the standard basis after a controlled Z on `(t,v)` is safe on both records, but measuring it before `v` controls a flip of `t` is unsafe on both. Both drop `t` and return `v`, inducing the same unread standard-basis measurement with weight $1/2$ for each record at every input.

4 Blocks and loans

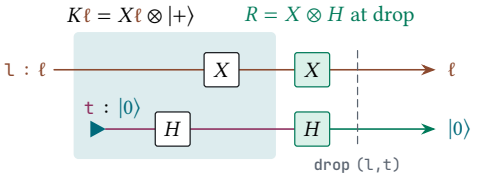
An allocated block in a fixed pure state is safe to drop even when its parts are mixed, as with a fresh Bell pair. We also consider blocks containing *loans*, registers borrowed in unknown states, possibly entangled with their caller, who expects them back unchanged.

The witness $\mathcal{S}(P,U)$ receives the unknown loan state ℓ, so its block factor is an isometry B rather than one fixed state g. A drop that names loans is safe when $K = B \otimes W$, with B acting on the loans and U computing W on the other inputs x.



With measurement, we again check each record o for $K_o = B_o \otimes W_o$, allowing R to depend on o. Without measurement, R restores the loan's normalized joint state with its caller reference, and the same holds on each occurring record when loan and reference are uncorrelated with other inputs. Otherwise measuring those inputs can change that state. The diagram below gives K and R for `lend`, whose unit return is `() : 1`.

```
prog lend (loan l : [2]) : 1 {
  new t : [2] @ |0>;
  H t; X l;
  drop (l,t); return () }
```



5 Scope and further work

We prove progress and preservation for a linear, first-order abstract machine with lifting and classical guards. For reuse, it restores safe blocks and pools temporaries for `new` of matching dimension and state, preserving records, weights and the reduced state of returned registers on each record. To decide safety, our exact Haskell checker tests each record's basis inputs for a common temporary state or loan isometry B, taking time exponential in the number of qubits. With only basis allocations and X, `CX` and Toffoli, safety of drops without loans is coNP-complete, even without measurement and with one drop (compare [11, Cor. 3.1]).

Our OpenQASM 3 compiler runs this check, computes numerical return gates, and reuses blocks safe on every record. Only basis allocations compile, program (c)'s V is unsupported, and no loan may be dropped after a `lift` or an `if`. For the general loan returns it supports, the compiler checks $RB = J$ entrywise within 10^{-9}. Measuring and resetting a qubit to $|0\rangle$ also permits reuse [4, 9], but destroys an unknown loan state.

Where Qurts and Reqomp reclaim by uncomputing [7, 15], our test constructs a return from the state at the drop, certifying the balanced Deutsch–Jozsa temporary that Qutes does not reclaim [5, Defs. 2–6]. Twist verifies annotated purity through static analysis and runtime separability tests [22, §1.3], but one return gate needs a pure state common to all inputs. The gate built from that state provides the restoration that ReQWIRE verifies [16, §4] and that Q# and fractional types require [3, 13].

For loans, local rotations repair separable errors on a dirty ancilla [10, §5.2, Alg. 2], while our gate undoes the block's isometry B at the drop. QBorrow also requires unchanged loans after each denoted operation [18, Def. 5.1] and Bona after each borrowing body [21, Def. 3.2], a handback property that separation logics verify deductively [12, 19].

Our language excludes loops and quantum recursion, so we leave open when a temporary used in a loop or recursive call is safe to drop. We also allow all unitaries as gates, although a fixed set may lack the return. For example, circuits over X, `CX`, Toffoli and $H \otimes H$ have dyadic entries [1, Cor. 5.6]. Applying $H \otimes H$ to `(t,v)` with `t` allocated in $|0\rangle$ gives a safe drop, yet restoring `t` alone requires non-dyadic entries $\pm 1/\sqrt{2}$.

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